

## Assignment 3

### Set A

**1. Write a program to read a string and copy it to another string and display copied string Also the length of copied string. (Use built in functions)**

**Program:**

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[10], str2[10];
    printf("Enter String: ");
    gets(str1);

    strcpy(str2, str1);
    printf("Copied String is %s\n", str2);
    printf("Length of the String is %d\n", strlen(str2));

    return 0;
}
```

**2. Write a program to read two strings. If first string is greater than second then concatenate second to first and display concatenated string, If first string is smaller than second then concatenate first to second, other-wise display length of string. (use strcmp)**

**Program:**

```
int main()
{
    char str1[10], str2[10];
    printf("Enter first string: ");
    gets(str1);
    printf("Enter second string: ");
    gets(str2);

    if(strcmp(str1, str2) > 0)
    {
        strcat(str1, str2);
        printf("String is %s\n", str1);
    }
    else
```

```

        if(strcmp(str1,str2)<0)
        {
            strcat(str2,str1);
            printf("String is %s\n",str2);
        }
        else
        {
            printf("Length of first string %s is : %d\n",str1,strlen(str1));
            printf("Length of second string %s is : %d\n",str2,strlen(str2));
        }

        return 0;
    }

```

**3. Write a program to read a string and one character. Check whether given character is present in the given string or not? (Hint: use strchr or strchr)**

```

#include <stdio.h>
#include<string.h>
int main()
{

    char str1[10];
    char ch,*p;
    printf("Enter Character : ");
    scanf("%c",&ch);
    printf("Enter first string: ");
    gets(str1);

    p=strchr(str1,ch);
    if(p==NULL)
    {
        printf("Character %c is not present in given String %s\n",*p,str1);
    }
    else
    {
        printf("Character %c is present in given String %s\n",*p,str1);
    }
    return 0;
}

```

Output

Enter Character : e

Enter first string: hello

Character e is present in given String hello

## Set B

1. Write a program which accepts a sentence from the user and alters it as follows: Every space is replaced by \*, case of all alphabets is reversed, digits are replaced by ?.

**Program :**

```
#include<stdio.h>
#include<ctype.h>
#include<string.h>

int main()
{
    char str[20];
    int i=0;
    printf("Enter String: ");
    fgets(str,20,stdin);
    while(str[i]!='\0')
    {
        if(str[i]==' ')
            str[i]='*';
        else
            if(islower(str[i]))
                str[i]=toupper(str[i]);
            else
                if(isupper(str[i]))
                    str[i]=tolower(str[i]);
                else
                    if(isdigit(str[i]))
                        str[i]='?';
                    i++;
    }
    printf("%s",str);
}
```

Output:

Enter String: This is RMC 123  
tHIS\*IS\*rmc\*???

2. Write a program which accepts a sentence from the user. Find and display reverse of it. (Don't use any function).

```
#include<stdio.h>
int main()
{
    char str[20],t;
    int i=0,len=0,j;
```

```

printf("Enter String: ");
fgets(str,20,stdin);

while(str[i]!='\0')
{
    i++;
}
len=i-1;
i=0;
while(i<len)
{
    t=str[i];
    str[i]=str[len];
    str[len]=t;
    i++;
    len--;
}
printf("Reverse String is: %s",str);
return 0;
}

```

Output:

Enter String: Hello

Reverse String is: olleH

### 3. Write a program that accepts n words and outputs them in dictionary order.

```

#include <stdio.h>
#include<string.h>
int main()
{
    char str[20][20],temp[20];
    int n,i,j;
    printf("How many word you want to enter");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter %d word: ",i+1);
        gets(str[i]);
    }
    for (i = 0; i < n;i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }
}

```

```

    }
}

}
printf("All the Words in Dictionary order : \n");
for(i=0;i<n;i++)
{
    printf("%s\n",str[i]);
}
return 0;
}

```

**4. Define two constant arrays of strings, one containing country names (ex: India, France(etc) and the other containing their capitals.(ex:Delhi,Paris etc).Note that country names and capital names have a one-one correspondence. Accept a country name from the user and display its capital. Example: Input: India , Output: Delhi.**

```

#include <stdio.h>
#include<string.h>
int main()
{
    char name[20],cname[10][20]={"India","France","Nepal","Pakistan","Germany" };
    char capital[10][20]={"Delhi","Paris","Kathmandu","Islamabad","Berlin"};
    int flag,i,j;

    for(i=0;i<5;i++)
    {
        printf("%s\n",cname[i]);
    }
    printf("Enter the Country name whose Capital you want: ");
    scanf("%s",name);
    for(i=0;i<5;i++)
    {
        if(strcmp(cname[i],name)==0)
        {
            printf("Capital of %s is: %s",cname[i],capital[i]);
            flag=0;
            break;
        }
        else
        {
            flag=1;
        }
    }

    if(flag==1)
    {
        printf("Invalid Country name");
    }
}

```

```
    return 0;  
}
```

**Output :**

India

France

Nepal

Pakistan

Germany

**Enter the Country name whose Capital you want:** Nepal

Capital of Nepal is: Kathmandu